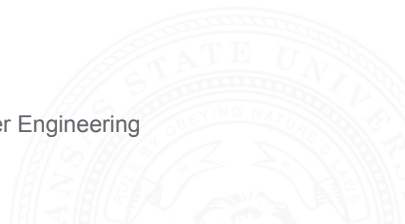


Three-Phase Transformers II

Three-phase network models, autotransformers, and taps

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Lecture outline

- 1 Build a balanced three-phase model from the single-phase equivalent.
- 2 A complete bank-rating and impedance example.
- 3 Autotransformer construction and operating concept.
- 4 Physical taps, off-nominal ratio, and model conventions.

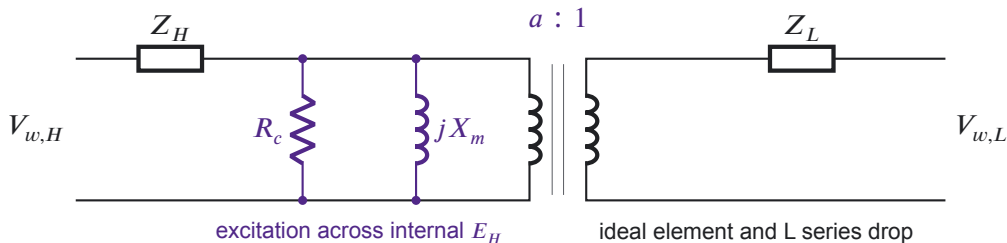
Establish actual H/L relations, choose compatible bases, then normalize.

PART 1

From single-phase to three-phase model

Quick recap: single-phase transformer

Single-phase unit: V_w is coil terminal voltage; $a = N_H/N_L$.



Three impedance parts

H series: $Z_H = R_H + jX_{\sigma,H}$.

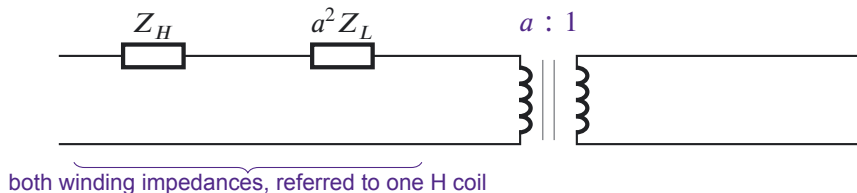
L series: $Z_L = R_L + jX_{\sigma,L}$.

Excitation: $R_c \parallel jX_m$ (core loss and magnetization).

Two winding series impedances, an excitation branch, and an ideal turns-ratio element.

Quick recap: reduced series model

Loaded operation: neglect I_0 when it is small relative to load current.



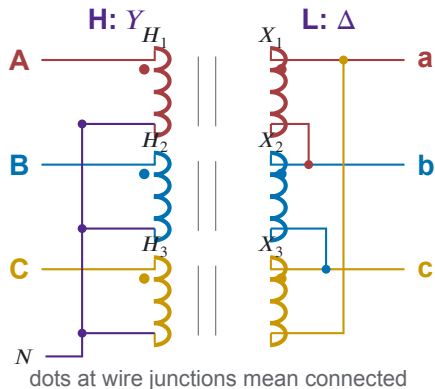
$$Z_H + a^2 Z_L = (R_H + a^2 R_L) + j(X_{\sigma,H} + a^2 X_{\sigma,L})$$

This is the reduced equivalent of one single-phase unit.

Retain $R_c \parallel jX_m$ when excitation or core loss is required.

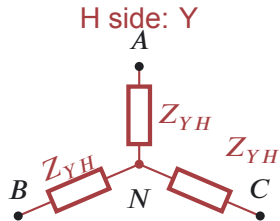
First choose the approximation; then refer and combine the series impedances.

Three-phase Y- Δ connection

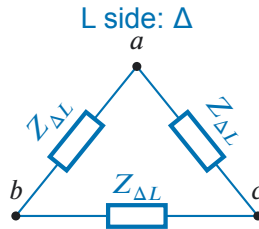


Series impedance on H or L

Balanced model: retain resistance and leakage; neglect excitation.



$$Z_{YH} = Z_H + a^2 Z_L$$



$$Z_{\Delta L} = Z_L + Z_H/a^2$$

Combined series impedances (Ω): H or L referral

Next: replace the L Δ impedance network with an equivalent Y.

L-side $\Delta \rightarrow Y$ equivalent



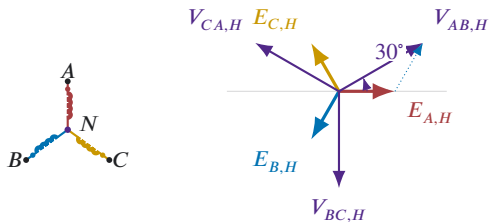
$$Z_{YL} = \frac{Z_{\Delta L}}{3}$$

n : equivalent L neutral reference; no grounding implied.

The impedance representation changes; the transformer stays Y- Δ .

Phase voltages and displacement

H: Y connection

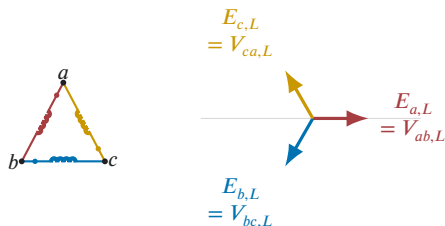


$$V_{AB,H} = \sqrt{3}e^{j30^\circ} E_{A,H}$$

$$\frac{V_{ab,L}}{V_{AB,H}} = \frac{1}{\sqrt{3}a} e^{-j30^\circ},$$

$$\delta_{LH} = \angle V_{ab,L} - \angle V_{AB,H} = -30^\circ$$

L: Delta abc connection



$$V_{ab,L} = E_{a,L} = E_{A,H}/a$$

Paired coil angles match. Y adds 30° , Delta abc adds 0° ; H line voltage leads L by 30° .

Actual terminal-phase voltages

Convert both line voltages to phases

$$V_{AN,H} = \frac{V_{AB,H}}{\sqrt{3}e^{j30^\circ}}$$
$$V_{an,L} = \frac{V_{ab,L}}{\sqrt{3}e^{j30^\circ}}$$

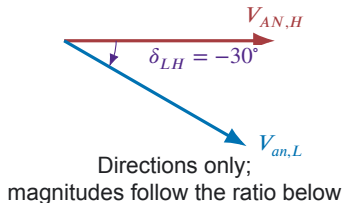
N : physical H neutral.

n : equivalent L reference; winding stays Delta.

$$V_{AN,H} = \sqrt{3}a e^{-j\delta_{LH}} V_{an,L}$$

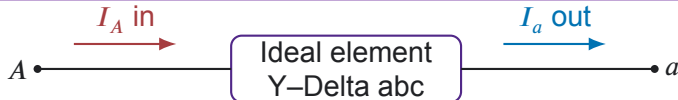
Actual RMS voltages; ideal nominal element, series drop omitted here.

The angle difference is unchanged



The ideal terminal-phase ratio has the same magnitude and angle as the line-voltage ratio.

Actual terminal currents



Terminal current references; phases B/b and C/c omitted

$$3V_{AN,H}I_A^* = 3V_{an,L}I_a^*$$

$$V_{AN,H} = \sqrt{3}a e^{-j\delta_{LH}} V_{an,L} \implies I_a^* = \sqrt{3}a e^{-j\delta_{LH}} I_A^*$$

$$I_a = \sqrt{3}a e^{j\delta_{LH}} I_A, \quad \frac{|I_A|}{|I_a|} = \frac{1}{\sqrt{3}a}$$

Actual RMS line currents (A); I_a also carries the equivalent L phase branch.

Ideal power balance sets the inverse current-magnitude ratio and conjugate angle factor.

Actual impedance and branch equation

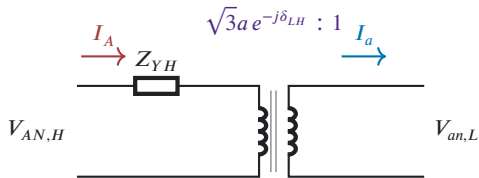
The same series effects, in ohms

$$Z_{YH} = Z_H + a^2 Z_L$$

$$Z_{YL} = \frac{Z_L + Z_H/a^2}{3}$$

$$Z_{YH} = 3a^2 Z_{YL}$$

$$V_{AN,H} = Z_{YH} I_A + \sqrt{3}a e^{-j\delta_{LH}} V_{an,L}, \quad I_a = \sqrt{3}a e^{j\delta_{LH}} I_A$$



Terminal-phase equivalent; excitation neglected.

Actual voltage and current ratios are known; the impedance ratio is their magnitude factor squared.

Define all system bases

Choose common three-phase S_B and line-voltage $V_{B,H}$

$$V_{B,L} = V_{B,H}/(\sqrt{3}a) \quad (\text{rated terminal ratio})$$

Derived base	H side	L side
Phase power	$S_B/3$	$S_B/3$
Terminal-phase voltage	$V_{B,\phi,H} = V_{B,H}/\sqrt{3}$	$V_{B,\phi,L} = V_{B,L}/\sqrt{3}$
Line current	$I_{B,H} = \frac{S_B}{\sqrt{3}V_{B,H}}$	$I_{B,L} = \frac{S_B}{\sqrt{3}V_{B,L}}$
Phase impedance	$Z_{B,H} = \frac{V_{B,H}^2}{S_B}$	$Z_{B,L} = \frac{V_{B,L}^2}{S_B}$

All bases are positive real magnitudes: MVA, kV, kA, and Ω with consistent units.

Choose S and one line-voltage base; the connection fixes the other side, then all phase bases follow.

Relations between H and L bases

Substitute the base definitions before normalizing any actual value

Base quantity	Relation between H and L
Voltage	$\frac{V_{B,\phi,H}}{V_{B,\phi,L}} = \frac{V_{B,H}}{V_{B,L}} = \sqrt{3}a$
Current	$\frac{I_{B,L}}{I_{B,H}} = \frac{V_{B,H}}{V_{B,L}} = \sqrt{3}a$
Impedance	$\frac{Z_{B,H}}{Z_{B,L}} = \left(\frac{V_{B,H}}{V_{B,L}} \right)^2 = 3a^2$

Shared $S_B = 3V_{B,\phi,H}I_{B,H} = 3V_{B,\phi,L}I_{B,L}$. Current bases therefore vary inversely with voltage.

The base ratios match the actual magnitude/referral factors; they carry no phase angle.

Normalize voltage and current

Voltage: divide actual by phase base

$$v_H = \frac{V_{AN,H}}{V_{B,\phi,H}}, \quad v_L = \frac{V_{an,L}}{V_{B,\phi,L}}$$

$$v_H = \frac{\sqrt{3}a e^{-j\delta_{LH}} V_{an,L}}{\sqrt{3}a V_{B,\phi,L}}$$

$$= e^{-j\delta_{LH}} v_L$$

Ideal nominal element; $\delta_{LH} = -30^\circ$. H current enters; L current leaves.

Current: divide actual by line base

$$i_H = \frac{I_A}{I_{B,H}}, \quad i_L = \frac{I_a}{I_{B,L}}$$

$$i_L = \frac{\sqrt{3}a e^{j\delta_{LH}} I_A}{\sqrt{3}a I_{B,H}}$$

$$= e^{j\delta_{LH}} i_H$$

The same magnitude factor cancels in each numerator and base; the connection angle remains.

Normalize impedance and power

Impedance: the H/L factors cancel

$$\begin{aligned} z_{eq} &= \frac{Z_{YH}}{Z_{B,H}} \\ &= \frac{3a^2 Z_{YL}}{3a^2 Z_{B,L}} = \frac{Z_{YL}}{Z_{B,L}} \end{aligned}$$

Same combined series impedance in pu on either side.

Power: use the common total MVA base

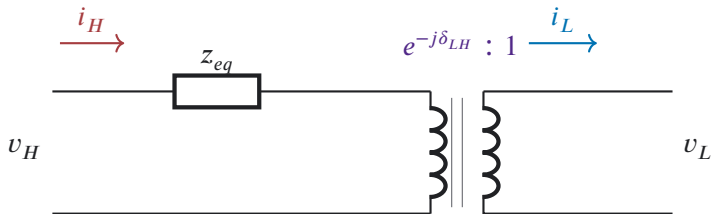
$$\begin{aligned} s &= \frac{S_{3\phi}}{S_B} = \frac{3V_{\phi} I_{\text{line}}^*}{3V_{B,\phi} I_B} \\ &= v i^* \end{aligned}$$

$$s_H = v_H i_H^*, \quad s_L = v_L i_L^*$$

Ideal element: $s_H = s_L$.

Compatible bases give one per-unit series impedance and remove the factor 3 from normalized power.

Assemble the balanced Y-Δ branch



balanced positive sequence; excitation neglected; resistance and leakage retained

$$v_H = \underbrace{z_{eq} i_H}_{\text{series drop}} + \underbrace{e^{-j\delta_{LH}} v_L}_{\text{ideal connection}}, \quad i_L = e^{j\delta_{LH}} i_H$$

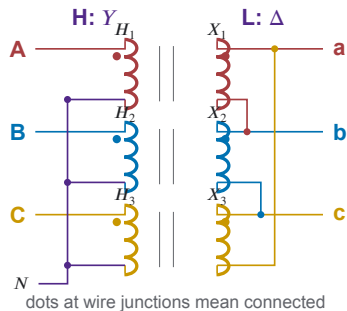
$$z_{eq} = r_{eq} + jx_{eq}; \delta_{LH} = -30^\circ; s_H - s_L = z_{eq} |i_H|^2 \text{ (all in pu).}$$

The final branch retains both winding series effects and the connection displacement.

PART 2

Bank example and checks

Bank example: data



Equipment and system bases

60 MVA, 138/13.8 kV, Y-Δ.

$$z_{60} = 0.006 + j0.090 \text{ pu}$$

$$S_B = 100 \text{ MVA}$$

$$V_{B,H} = 138 \text{ kV}$$

$$V_{B,L} = 13.8 \text{ kV}$$

Paired coils: (AN, ab), (BN, bc), (CN, ca).

Positive sequence; H neutral N available.

Grounding method is unspecified.

The kV ratings are line-to-line voltages; 60 MVA is the total bank rating.

Bank example: solution route

- | | |
|-----------------------|---|
| 1. Given | Rated terminal kV, bank MVA, coil joining, and series-impedance data. |
| 2. Model | Balanced positive-sequence branch; excitation neglected, resistance and leakage retained. |
| 3. Calculation | Recover coil ratio; derive L phasor; convert the combined series impedance. |
| 4. Check | Check actual ohms and rated currents independently. |

Use physical quantities to check the normalized model.

Bank example: voltage ratios

Coil voltages

$$E_H = 138/\sqrt{3} = 79.6743 \text{ kV}$$

$$E_L = 13.8 \text{ kV}$$

$$a = 10/\sqrt{3} = 5.77350$$

Terminal voltages

$$\frac{V_{LL,H,\text{rated}}}{V_{LL,L,\text{rated}}} = \frac{138}{13.8} = 10$$

$$\frac{V_{LL,H,\text{rated}}}{V_{LL,L,\text{rated}}} = \sqrt{3}a$$

Both ratios describe the same bank.

The Y-Δ coil ratio is the terminal line ratio divided by $\sqrt{3}$.

Bank example: no-load phasor

Reference: $V_{AB,H} = 138\angle 0^\circ$ kV.

$$V_{AN,H} = \frac{V_{AB,H}}{\sqrt{3}e^{j30^\circ}} = 79.6743\angle(-30^\circ) \text{ kV}$$

$$\begin{aligned} V_{ab,L} &= V_{AN,H}/a \\ &= 11.95115 - j6.90000 = 13.800\angle(-30^\circ) \text{ kV} \end{aligned}$$

$$V_{an,L} = \frac{V_{ab,L}}{\sqrt{3}e^{j30^\circ}} = 7.96743\angle(-60^\circ) \text{ kV}$$

$V_{an,L}$ uses the equivalent L neutral reference.

Transfer the AN–ab coil voltage, then form the equivalent L phase voltage.

Bank example: change of base

$$z_{new} = z_{old} \frac{S_{B,new}}{S_{B,old}} \left(\frac{V_{B,old}}{V_{B,new}} \right)^2$$

The voltage bases on each physical side are unchanged.

$$z_{100} = (0.006 + j0.090) \frac{100}{60} = 0.010 + j0.150 \text{ pu}$$

The per-unit number changes because the impedance base changes; the physical impedance stays fixed.

Bank example: recover ohms

Side	$Z_{B,60} (\Omega)$	$Z_Y (\Omega)$
H (Y)	$138^2/60 = 317.4$	$1.9044 + j28.566$
L (equiv. Y)	$13.8^2/60 = 3.174$	$0.019044 + j0.28566$

$$\frac{Z_{YH}}{Z_{YL}} = \left(\frac{V_{LL,H,\text{rated}}}{V_{LL,L,\text{rated}}} \right)^2 = 100$$

On the 100-MVA system base, $Z_{B,H} = 190.44 \Omega$ and $Z_{B,L} = 1.9044 \Omega$.

L Δ branches: $Z_{\Delta L} = 3Z_{YL} = 0.057132 + j0.85698 \Omega$.

Either MVA base reproduces the same physical terminal equivalent.

Bank example: rated currents

Equipment currents

$$I_{H,r} = \frac{60}{\sqrt{3}(138)} = 0.251022 \text{ kA}$$

$$I_{L,r} = \frac{60}{\sqrt{3}(13.8)} = 2.51022 \text{ kA}$$

Coil-current magnitudes:

$$I_{w,H} = I_{H,r} \text{ (Y).}$$

$$I_{w,L} = I_{L,r}/\sqrt{3} = 1.44928 \text{ kA } (\Delta).$$

System-base check

$$I_{H,B} = 0.41837 \text{ kA}$$

$$I_{L,B} = 4.18370 \text{ kA}$$

Rated current is 0.60 pu on the 100-MVA base.

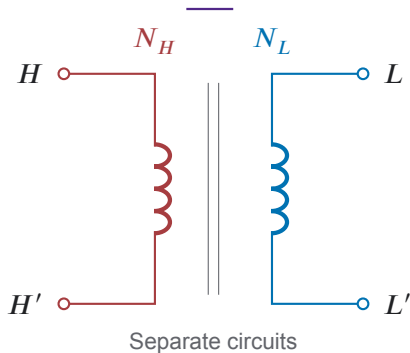
Rated equipment current is 1 pu on its own MVA base and 0.60 pu on the system base.

PART 3

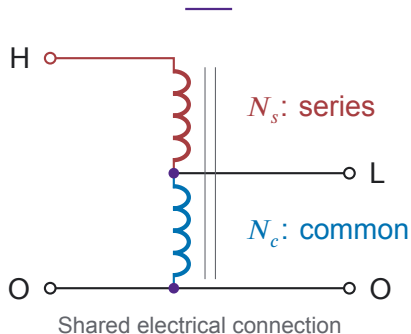
Autotransformer concept

Two-winding and autotransformer

Two-winding transformer

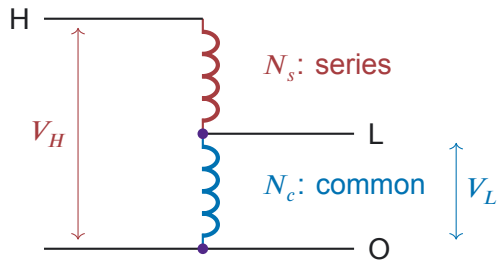


Autotransformer



An autotransformer couples H and L electrically as well as magnetically.

Series and common sections



Series-aiding connection

H is connected across $N_s + N_c$ turns. L is connected across N_c turns.

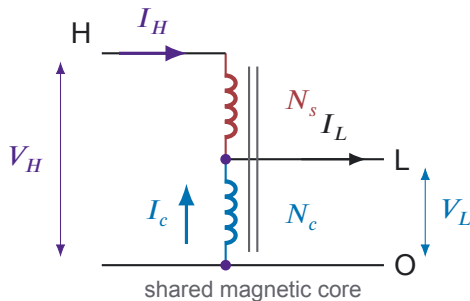
$$V_H = E_s + E_c, \quad V_L = E_c$$

$$\frac{V_H}{V_L} = \frac{N_s + N_c}{N_c}$$

The common section belongs to both terminal circuits.

Source: Glover et al., sixth edition (2017), Fig. 3.23(b)

Autotransformer power paths



$$I_L = I_H + I_c \quad (I_c : O \rightarrow L)$$

Ideal step-down; RMS phasors

S : terminal complex power

$$S = V_H I_H^* = V_L I_L^*$$

Direct: shared electrical path

$$S_{\text{direct}} = V_L I_H^*$$

Magnetic: coupling through core

$$S_{\text{mag}} = V_L I_c^* = (V_H - V_L) I_H^*$$

$$S = S_{\text{direct}} + S_{\text{mag}}$$

The common winding supplies I_c ; the direct current component is I_H .

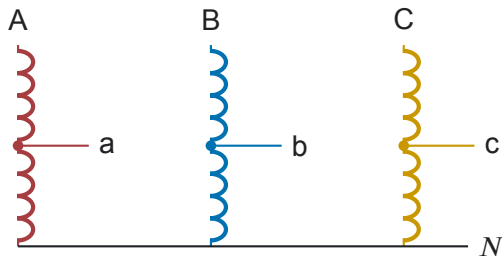
Source: Glover et al., sixth edition (2017), Fig. 3.23(b), Sec. 3.7

Autotransformer applications

- Applications** Voltage-level interconnection and voltage adjustment.
- Material saving** Nearby voltage levels give a smaller magnetic power share, allowing less winding material and a compact unit.
- Isolation** H and L share an electrical path; there is no galvanic isolation.
- System effects** Grounding and transferred overvoltages are linked. Lower impedance can increase short-circuit current.



Three-phase autotransformer



illustrated common-neutral Y-connected implementation

Each phase has a series section and a common section.

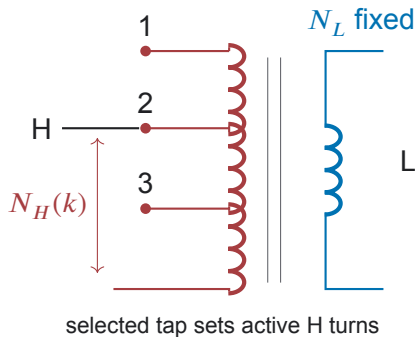
Three-phase construction repeats the autoconnected winding path on all three phases.

Source: Course-redrawn conceptual three-phase extension of textbook Fig. 3.23(b).

PART 4

Tap ratio concept

Tapped winding construction



A selected connection point

Changing position k changes the active turns $N_H(k)$.

$$a(k) = \frac{N_H(k)}{N_L}$$

The Y/ Δ connections remain as specified. A tapped isolated transformer still has separate H and L circuits.

A winding tap changes active turns without necessarily creating an autotransformer.

Source: Course-redrawn conceptual tap circuit. Reinhausen, Tap-changer basics, official description of active-turn selection.

Three different ratios

- 1 Physical winding ratio: $a(k) = N_H(k)/N_L$; k labels tap position.
- 2 Actual line-voltage ratio: $|V_{LL,H}(k)|/|V_{LL,L}(k)|$, from $a(k)$ and Y/ Δ joining.
- 3 Voltage-base ratio: $b = V_{B,H}/V_{B,L}$, fixed by the system model.

$$\tau(k) = \frac{|V_{LL,H}(k)|/|V_{LL,L}(k)|}{V_{B,H}/V_{B,L}}$$

With bases in the rated line-voltage ratio, the nominal tap has $\tau = 1$.

The off-nominal per-unit tap is actual terminal ratio divided by the chosen base ratio.

Tap-changing equipment

- De-energized** Change tap position with the transformer de-energized. Select a planned operating ratio.
- On-load** Change tap position while supplying load. Adjust the ratio during operation.
- Transition path** On-load switching maintains the load-current path. A transition resistor or reactor limits current between adjacent taps.



Summary

- 1 Choose the model level; retain winding resistance and leakage, and state the excitation assumption.
- 2 Use the connection to obtain phase impedance; retain voltage and current displacement.
- 3 An autotransformer has shared series/common sections and no H/L galvanic isolation.
- 4 An off-nominal tap is the actual ratio divided by the chosen base ratio.

A useful transformer model preserves the physical connection and names every ratio.